

Forecast band calibration: NL system load

A worked example of the deliverable, formatted exactly as a client receives it. It is run on a public series rather than on client data, and every number in it regenerates from the linked script and notebook.

SUBJECT	PERIOD
Day-ahead system load forecast, Netherlands (ENTSO-E)	2025-07 to 2026-07
SCORED	BASIS
38,016 quarter-hours, 336 days after warm-up	Specimen on public data

Verdict

The two bands a desk builds without a calibration step do not deliver the coverage they state, and they fail hardest in the quarter-hours where being wrong is most expensive. Of 3 constructions audited, 2 carry Material findings and 1 does not fail on this evidence.

ID	FINDING	SEVERITY	EXPOSURE AT 90%
F-01	Parametric Gaussian band at nominal 90% covered 84.2%	MATERIAL	EUR 362,495
F-02	Rolling split-conformal band at nominal 90% covered 82.1%	MATERIAL	EUR 359,943
F-03	Adaptive conformal band at nominal 90% covered 88.6%	OBSERVATION	EUR 158,145

Scope and exclusions

What an audit did not look at is the half a reader cannot infer, so it is stated rather than left to the method section.

WHAT THIS AUDIT DID NOT CHECK

- Whether the forecast itself can be improved. This audits the stated uncertainty around it, not the point forecast.
- Any series other than system load, and any period outside the 13 months above.
- Whether the desk actually reserves flexibility equal to the stated band. The exposure figures assume it does; if it reserves more, they overstate.
- Price risk beyond the imbalance-versus-day-ahead spread. No intraday trading, no portfolio effects, no cross-commodity exposure.
- Whether the adaptive construction would hold on a future regime change. It is reactive by design and this report says so under F-03.

Findings register

F-01

MATERIAL

Parametric Gaussian band at nominal 90% covered 84.2%

WHY THIS SEVERITY

nominal outside the coverage interval, and misses run 1.36x more frequent in the most expensive intervals (interval lower bound $1.14 > 1$)

MEASUREMENT

Headline 90%: coverage 0.842, day-block bootstrap 95% interval [0.816, 0.868] over 32,252 scored intervals. Every audited level: 80%: covered 74.4% [71.3%, 77.4%], concentration 1.29x [1.13, 1.47], Material; 90%: covered 84.2% [81.6%, 86.8%], concentration 1.36x [1.14, 1.59], Material; 95%: covered 89.6% [87.3%, 91.8%], concentration 1.43x [1.15, 1.76], Material

CONSEQUENCE

EUR 362,495 of un-reserved adverse exposure per 100 MW mean load over the audited period at the 90% level

RECOMMENDED ACTION

Re-state the band at its measured coverage, or recalibrate. The adaptive construction audited here is one option and its width cost is quantified.

LIMIT OF THIS FINDING

Measured on this series over this period only. The interval is block-bootstrapped by day and so already carries within-day dependence; it does not carry uncertainty in the price series.

F-02

MATERIAL

Rolling split-conformal band at nominal 90% covered 82.1%

WHY THIS SEVERITY

nominal outside the coverage interval, and misses run 1.31x more frequent in the most expensive intervals (interval lower bound $1.13 > 1$)

MEASUREMENT

Headline 90%: coverage 0.821, day-block bootstrap 95% interval [0.793, 0.847] over 32,252 scored intervals. Every audited level: 80%: covered 71.5% [68.3%, 74.8%], concentration 1.25x [1.10, 1.40], Significant; 90%: covered 82.1% [79.3%, 84.7%], concentration 1.31x [1.13, 1.53], Material; 95%: covered 88.6% [86.2%, 90.9%], concentration 1.36x [1.09, 1.65], Material

CONSEQUENCE

EUR 359,943 of un-reserved adverse exposure per 100 MW mean load over the audited period at the 90% level

RECOMMENDED ACTION

Re-state the band at its measured coverage, or recalibrate. The adaptive construction audited here is one option and its width cost is quantified.

LIMIT OF THIS FINDING

Measured on this series over this period only. The interval is block-bootstrapped by day and so already carries within-day dependence; it does not carry uncertainty in the price series.

F-03

OBSERVATION

Adaptive conformal band at nominal 90% covered 88.6%

WHY THIS SEVERITY

nominal coverage lies inside the day-block bootstrap interval; this evidence does not show the band failing at this level

MEASUREMENT

Headline 90%: coverage 0.886, day-block bootstrap 95% interval [0.865, 0.906] over 32,252 scored intervals. Every audited level: 80%: covered 77.6% [74.8%, 80.3%], concentration 1.19x [1.04, 1.37], Observation; 90%: covered 88.6% [86.5%, 90.6%], concentration 1.26x [1.03, 1.52], Observation; 95%: covered 93.6% [92.0%, 95.1%], concentration 1.15x [0.89, 1.48], Observation

CONSEQUENCE

EUR 158,145 of un-reserved adverse exposure per 100 MW mean load over the audited period at the 90% level

RECOMMENDED ACTION

Re-state the band at its measured coverage, or recalibrate. The adaptive construction audited here is one option and its width cost is quantified.

LIMIT OF THIS FINDING

Measured on this series over this period only. The interval is block-bootstrapped by day and so already carries within-day dependence; it does not carry uncertainty in the price series.

Evidence

Coverage against the level each band states

Intervals are 95% day-block bootstrap, resampling whole days. A band is only recorded as failing when the level it states falls outside that interval.

CONSTRUCTION	STATES	DELIVERED	95% INTERVAL	VERDICT
Parametric Gaussian	80.0%	74.4%	[71.3%, 77.4%]	fails
Parametric Gaussian	90.0%	84.2%	[81.6%, 86.8%]	fails
Parametric Gaussian	95.0%	89.6%	[87.3%, 91.8%]	fails
Rolling split-conformal	80.0%	71.5%	[68.3%, 74.8%]	fails
Rolling split-conformal	90.0%	82.1%	[79.3%, 84.7%]	fails
Rolling split-conformal	95.0%	88.6%	[86.2%, 90.9%]	fails
Adaptive conformal	80.0%	77.6%	[74.8%, 80.3%]	holds
Adaptive conformal	90.0%	88.6%	[86.5%, 90.6%]	holds
Adaptive conformal	95.0%	93.6%	[92.0%, 95.1%]	holds

Where the misses fall

An interval is *expensive* when its adverse imbalance spread lands in the top 20% of all scored intervals, which is EUR 42.29/MWh here. That threshold was fixed before any band was scored. The ratio below is the miss rate inside those intervals against the miss rate everywhere else.

CONSTRUCTION	STATES	MISS RATE, EXPENSIVE	MISS RATE, REST	RATIO	95% INTERVAL
Parametric Gaussian	80.0%	31.2%	24.1%	1.29x	[1.13, 1.47]
Parametric Gaussian	90.0%	20.1%	14.7%	1.36x	[1.14, 1.59]
Parametric Gaussian	95.0%	13.6%	9.5%	1.43x	[1.15, 1.76]
Rolling split-conformal	80.0%	33.9%	27.2%	1.25x	[1.10, 1.40]
Rolling split-conformal	90.0%	22.2%	16.9%	1.31x	[1.13, 1.53]
Rolling split-conformal	95.0%	14.4%	10.6%	1.36x	[1.09, 1.65]
Adaptive conformal	80.0%	25.7%	21.6%	1.19x	[1.04, 1.37]
Adaptive conformal	90.0%	13.7%	10.9%	1.26x	[1.03, 1.52]
Adaptive conformal	95.0%	7.2%	6.3%	1.15x	[0.89, 1.48]

What the bands cost

For every quarter-hour the outcome escaped the stated band, the excess beyond the band edge is exposure the band said not to carry. It is priced at that interval's adverse imbalance spread and scaled to a 100 MW mean-load portfolio.

CONSTRUCTION	80%	90%	95%
Parametric Gaussian	EUR 586,617	EUR 362,495	EUR 241,799
Rolling split-conformal	EUR 656,087	EUR 359,943	EUR 207,305
Adaptive conformal	EUR 396,290	EUR 158,145	EUR 78,539

The mean adverse spread across all scored intervals was EUR 31.0/MWh. Across the intervals the Gaussian 90% band missed it was EUR 39.4/MWh. **The band fails disproportionately in the intervals where failing is expensive.**

Conditional bias, by delivery slot

Reported as a diagnostic, not a scored finding: no severity threshold was pre-registered for bias, and inventing one after seeing the result is exactly what the register is built to avoid. The largest slot-mean bias is 3,854 MW at 13:00 local, and it runs the opposite way to the headline: the forecast is too low at that slot.

What a band costs in width

CONSTRUCTION	STATES	MEAN WIDTH	95TH PCT	WIDEST
Parametric Gaussian	80.0%	3,019 MW	5,347 MW	8,601 MW
Parametric Gaussian	90.0%	3,875 MW	6,863 MW	11,039 MW
Parametric Gaussian	95.0%	4,617 MW	8,178 MW	13,154 MW
Rolling split-conformal	80.0%	2,831 MW	5,267 MW	9,118 MW
Rolling split-conformal	90.0%	3,720 MW	6,781 MW	11,839 MW
Rolling split-conformal	95.0%	4,581 MW	8,087 MW	13,489 MW
Adaptive conformal	80.0%	3,473 MW	6,939 MW	15,109 MW
Adaptive conformal	90.0%	4,730 MW	8,759 MW	15,940 MW
Adaptive conformal	95.0%	5,607 MW	9,789 MW	15,940 MW

Assumptions register

Every number above rests on these. The direction column says which way the headline moves if the assumption is wrong, so a reader can discount it without having to guess.

ASSUMPTION	IF IT IS WRONG
Portfolio errors move proportionally to system errors, scaled to 100 MW mean load	Either direction, and this is the single largest lever on the exposure figures. A portfolio less correlated with the system carries less exposure than stated; a concentrated one carries more.
Favourable spreads clipped to zero, risk side only	Inflates the exposure figures. A desk that earns on favourable imbalance would net some of this back.
The desk reserves flexibility equal to the stated band	Inflates if the desk already reserves more than it states.
Trailing 90-day window, 60-day minimum, no lookahead	A longer window smooths regime breaks and would flatter the static constructions; a shorter one is noisier. Fixed before scoring.
Adaptive step size $\gamma = 0.02$	Fixed before evaluation, not tuned to this series. A larger step reacts faster and quotes wider.
Imbalance and day-ahead prices as published, unrevised	Revisions would move the exposure figures, not the coverage findings.

Method note

How severity is assigned

Severity is derived by rule, not asserted. The rule is fixed in the script and applied identically to every construction and level.

MATERIAL

The stated level falls outside the day-block bootstrap interval, *and* misses concentrate in expensive intervals by at least 1.25x, *and* that concentration interval excludes 1.

SIGNIFICANT

The stated level falls outside the interval, but the concentration test fails either limb: not distinguishable from none, or real but below the materiality floor.

OBSERVATION

The stated level falls inside the interval. The band holds at this level on this evidence, and sample size is allowed to win.

A construction takes the most severe rating it reaches at any audited level. The 1.25x floor is a commercial judgement rather than a statistical one: it stops a concentration of 1.02 with a tight interval being called Material. It is stated here so it can be argued with.

THE THRESHOLD BINDS AGAINST THE REPORT

Rolling split-conformal at the 80% level has a concentration of **1.2499x** against a floor of 1.25. It misses Material by one ten-thousandth and is recorded as Significant. A threshold chosen after seeing the results would not have landed there.

Why the intervals are bootstrapped and not binomial

Quarter-hour forecast errors are autocorrelated within a day and across neighbouring slots. Treating 38,016 intervals as independent trials would give a binomial interval far tighter than the evidence supports, and would promote findings that sample size cannot carry. Every interval here resamples whole *days* with replacement, 2,000 times, seed 20260901. That is why all three adaptive findings score Observation rather than a small win.

Lineage

DATA	ENTSO-E Transparency Platform, zone NL: load forecast, load actual, day-ahead price, imbalance price
SCRIPT	src/audit_specimen.py -zone NL
COMMIT	04409d9
OUTPUTS	artifacts/specimen/: coverage.json, cost.json, findings.json, findings.csv, monthly.json, bias_slot.json, meta.json
REGISTER AS DATA	findings.csv carries all nine construction-level rows, including the levels this document summarises

A client engagement replaces the public series with your own and changes nothing else. The German equivalent of this report is the same script against reBAP.